1. Review of Riemann Integration

REAL VARIABLES I

Fall 2026

We briefly review the construction and properties of the Riemann integral and identify its shortcomings that suggest constructing a new integration theory: the Lebesgue integral.

First, we show that continuous functions are a natural class on which the Riemann integral is defined. However some discontinuities can be tolerated - we anticipate a result we prove later in the course that provides a complete characterization of Riemann integrable functions. However, this result is formulated in terms of quantities appearing in Lebesgue integration theory.

Second, we use the integral to measure the distance between functions and thereby introduce the norm L^1. We will show that continuous functions with the distance induced by the L^1 norm is not a complete metric space, thereby having the same issues for the purpose of analysis as the rationals have with respect to the reals.

Abstract metric space theory tells us that there exists a unique *completion* of the space of continuous functions, to which the Riemann integral can be extended to. This happens to be exactly the Lebesgue integral. This description is completely opaque, but it does provide an important goal: create an integration theory such that the associated spaces of integrable functions are complete.

1.1 Riemann integrability

A function is Riemann integrable if lower and upper Riemann sum approximations can be made arbitrarily close.

Definition 1.1.1 (Partition of an interval)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A **partition** of $[a, b]$ is a finite ordered tuple $\tau = (\tau_0, \tau_1, \dots, \tau_n)$ such that $a = \tau_0 < \tau_1 < \dots < \tau_n = b$.. The **mesh** of the partition is

$$|\tau| := \max_{0 \le j \le n-1} (x_{j+1} - x_j).$$

The number of intervals in the partition is denoted $\#\tau := n$.

Warning. The term *partition* is heavily overloaded in mathematics. Often it stands for an arbitrary representation of a set as a disjoint union. Here, we partition an interval into *finitely many* ordered intervals.

Definition 1.1.2 (Upper and lower Riemann sums)**.** The **upper Riemann sum** and **lower Riemann sum** of f with respect to τ are

$$U(f;\tau) := \sum_{j=0}^{\#\tau-1} \big(\sup_{x\in[\tau_j,\tau_{j+1}]} f(x)\big)(\tau_{j+1}-\tau_j)$$
$$L(f;\tau) := \sum_{j=0}^{\#\tau-1} \big(\inf_{x\in[\tau_j,\tau_{j+1}]} f(x)\big)(\tau_{j+1}-\tau_j).$$

The lower sum never exceeds the upper sum. Refining a partition can only raise the lower sum and lower the upper sum.

Definition 1.1.3 (Riemann integrability)**.** A bounded function $f : [a, b] \to \mathbb{R}$ is **Riemann integrable** if

$$\sup_{\tau} L(f;\tau) = \inf_{\tau} U(f;\tau),$$

where the supremum and infimum are taken over all finite partitions of $[a, b]$.

The common value is denoted by

$$\int_a^b f(x)\,dx.$$

Therefore, to show that a function is Riemann integrable it is sufficient to show that by choosing an appropriate partition τ, the quantity $U(f;\tau) - L(f;\tau)$ can be made arbitrarily small. This is also necessary because for any given $\varepsilon > 0$ if $U(f;\tau) - L(f;\sigma) < \varepsilon$ for two different partitions τ and σ then $U(f;\gamma) - L(f;\gamma) < \varepsilon$ is taken to be a common refinement of the partitions τ, σ.

According to this reduction, we must show that

$$U(f;\tau) - L(f;\tau) = \sum_{j=0}^{\#\tau-1} \operatorname{osc}(f;[\tau_j,\tau_{j+1}])\,(\tau_{j+1}-\tau_j),$$

can be made arbitrarily small, where the oscillation of f over an interval J is defined as

$$\operatorname{osc}(f;J) := \sup_{x\in J} f(x) - \inf_{x\in J} f(x).$$

We are ready to prove the following theorem.

Theorem 1.1.4 (Continuous functions are Riemann integrable)**.** Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann integrable.

Proof. (proves 1.1.4)

The claim follows by bounding (1.1). Since $f : [a, b] \to \mathbb{R}$ is continuous and $[a, b]$ is compact, it admits a modulus of continuity i.e. a function $\omega_f : [0, +\infty) \to [0, +\infty)$ such that

- $\omega_f(t) \to 0$ as $t \to 0^+$
- $|f(x_2) - f(x_1)| \leq \omega_f(|x_2 - x_1|)$ for any $x_1, x_2 \in [a, b]$.

We have the bound

$$\begin{aligned} U(f;\tau) - L(f;\tau) &= \sum_{j=0}^{\#\tau-1} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\,(\tau_{j+1} - \tau_j), \\ &\leq \sum_{j=0}^{\#\tau-1} \omega_f(|\tau|)(\tau_{j+1} - \tau_j) = \omega_f(|\tau|)(b - a). \end{aligned}$$

where we used that

$$\operatorname{osc}(f; [\tau_j, \tau_{j+1}]) \leq \omega_f(\tau_{j+1} - \tau_j) \leq \omega_f(|\tau|).$$

It is therefore sufficient to choose a partition τ such that $|\tau|$ and, therefore, $\omega_f(|\tau|)$ is as small as desired. $\square$

1.1.1 Digression on uniform continuity and modulus of continuity

The proof above essentially relies on the following reasoning.

Suppose every subinterval of a partition has length at most δ. If the values of f differ by at most η whenever their inputs are within δ of one another, then every oscillation in (1.1) is at most η, and therefore

$$U(f;\tau) - L(f;\tau) \leq \eta(b - a).$$

The point is that one value of δ must work everywhere on $[a, b]$, not merely near one point at a time.

Definition 1.1.5 (Uniform continuity)**.** Let $E \subseteq \mathbb{R}$ and let $f : E \to \mathbb{R}$. The function f is **uniformly continuous** on E if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in E$,

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

The number δ may depend on ε, but it must work for every pair $x, y \in E$.

For continuity at a point x_0, the choice of δ may depend on x_0. A function can therefore be continuous at every point while requiring smaller and smaller scales as the point moves through the domain. Uniform continuity rules this out by requiring one common scale.

Theorem (Review: Heine–Cantor theorem)**.** Let $K \subseteq \mathbb{R}$ be compact. Every continuous function $f : K \to \mathbb{R}$ is uniformly continuous on K. In particular, every continuous function on a closed bounded interval $[a, b]$ is uniformly continuous.

The proof is a prototypical compactness argument, which we omit as it is beyond the scope of this exposition, but we invite the reader to refresh their memory.

Generally, continuous functions are not uniformly continuous.

Problem. Let $\alpha > 0$. Show that $x \mapsto |x|^\alpha$ is continuous on $\mathbb{R}$ but it is uniformly continuous there only if $\alpha \leq 1$.

The modulus of continuity encodes the $\varepsilon - \delta$ relationship of Definition 1.1.5 by recording how much a function can change at each spatial scale.

Definition (Modulus of continuity)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. A **modulus of continuity** for f is a nondecreasing function

$$\omega : [0, \infty) \to [0, \infty)$$

such that

$$\omega(t) \longrightarrow 0 \qquad \text{as } t \to 0^+,$$

and

$$|f(x_2) - f(x_1)| \leq \omega(|x_2 - x_1|) \qquad \forall x_1, x_2 \in I.$$

The natural choice is the largest change of value of f over all pairs separated by at most δ:

$$\omega_f(\delta) := \sup\{|f(x_2) - f(x_1)| : x_1, x_2 \in I,\ |x_2 - x_1| \leq \delta\}.$$

We suggest the reader to recall and know how to prove the following fact.

Proposition (Review: Uniform continuity is equivalent to having a modulus)**.** Let $E \subseteq \mathbb{R}$ be an interval or a finite union of disjoint intervals and let $f : I \to \mathbb{R}$. Then f is uniformly continuous on E if and only if it has a modulus of continuity.

When f is uniformly continuous, the function ω_f in (1.1.1) is finite for every $\delta > 0$ and is a modulus of continuity.

We can now complete the argument that motivated uniform continuity.

1.1.2 Riemann integrability of functions with discontinuities

Continuity controlled the oscillation on every subinterval. If a bounded function has finitely many discontinuities, that control fails near those points. The proof can still succeed by treating two regions differently. Away from the discontinuities, continuity makes the oscillation small. Near the discontinuities, one uses boundedness of f and smallness of the intervals in τ containing the discontinuities.

Proposition 1.1.6 (Bounded functions with finitely many discontinuities are Riemann integrable)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. If f has only finitely many discontinuities, then f is Riemann integrable.

We provide a proof for completeness, but it is generally of little interest for this course and the reader is invited to skip it.

Proof. Let $D = \{d_1, \dots, d_N\}$ be the set of discontinuities. By assumption $\|f\|_{\sup} < \infty$.

Fix $r > 0$ sufficiently small and note that f is uniformly continuous on the finite union of intervals $[a, b] \setminus E_r$ with $E_r \coloneqq \setminus \bigcup_{j=1}^{N} B_r(d_j)$ being the exceptional set at scale r. Let ω_r be the modulus of continuity of f on $[a, b] \setminus E_r$.

Given any partition τ, let $\mathcal{J}_{\tau,r} := \left\{ j \in \{0, \dots, \#\tau\} : [\tau_j, \tau_{j+1}] \cap E_r \neq \emptyset \right\}$, that is the collection of indexes such that the interval $[\tau_j, \tau_{j+1}]$ intersects the exceptional set E_r.

Therefore, for any partition τ,

$$
\begin{aligned}
U(f;\tau) - L(f;\tau) = & \sum_{j \notin \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\, (\tau_{j+1} - \tau_j). \\
& + \sum_{j \in \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\, (\tau_{j+1} - \tau_j)
\end{aligned}
$$

For the first summand we have that

$$
\sum_{j \notin \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\, (\tau_{j+1} - \tau_j) \le \omega_r(|\tau|)(b - a)
$$

because all intervals in play are in $[a, b] \setminus E_r$.

For the second summand we use that $\operatorname{osc}(f; [\tau_j, \tau_{j+1}]) < 2\|f\|_{\sup}$ but that

$$
\bigcup_{j \in \mathcal{J}_{\tau,r}} [\tau_j, \tau_{j+1}] \subseteq E_{r+|\tau|}
$$

since the intervals $[\tau_j, \tau_{j+1}]$ touch the exceptional set E_r, consisting of r-neighborhoods of the points D, the intervals themselves have length at most $|\tau|$. Therefore

$$\sum_{j \in \mathcal{J}_{\tau,r}} (\tau_{j+1} - \tau_j) \leq N \cdot 2 \cdot (r + |\tau|)$$

and so

$$\sum_{j \in \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}]) \, (\tau_{j+1} - \tau_j) \leq 2\|f\|_{\sup} N \cdot 2 \cdot (r + |\tau|)$$

Combining the two estimates we obtain that

$$U(f;\tau) - L(f;\tau) \leq \omega_r(|\tau|)(b-a) + 2\|f\|_{\sup} N \cdot 2 \cdot (r + |\tau|).$$

By choosing τ with $|\tau|$ as small as required, we obtain that

$$\inf_\tau (U(f;\tau) - L(f;\tau)) \leq 2\|f\|_{\sup} N \cdot 2 \cdot r.$$

Now, since $r > 0$ was arbitrary we obtain that

$$\inf_\tau (U(f;\tau) - L(f;\tau)) = 0.$$

□

This proof shows that finiteness is not the essential issue. What mattered was the ability to cover all discontinuities by intervals whose total length is as small as desired. That observation leads to the exact characterization that will be proved later in the course.

Definition 1.1.7 (Lebesgue null set)**.** A set $E \subseteq \mathbb{R}$ has **Lebesgue measure zero**, or is a **Lebesgue null set**, if for every $\varepsilon > 0$ there exists a sequence of open intervals $(a_k, b_k)_{k \in \mathbb{N}}$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} (a_k, b_k)$$

and

$$\sum_{k=1}^{\infty} |b_k - a_k| < \varepsilon.$$

A null set need not be finite. The definition says that, from the point of view of total interval length, the set can be confined to a region of arbitrarily small ”length.”

Theorem 1.1.8 (Preview: Lebesgue's criterion for Riemann integrability)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded, and let

$$D_f := \{x \in [a, b] : f \text{ is discontinuous at } x\}.$$

Then f is Riemann integrable if and only if D_f has Lebesgue measure zero.

No proof is given here. The theorem states the destination of the first part of the course: Riemann integrability is governed not by the number of discontinuities, but by whether their locations can be covered by intervals of arbitrarily small total length. This last quantity is a prototypical construction in Lebesgue measure theory.

1.2 The L^1 norm and incompleteness

On $C([a, b])$, define

$$\|f\|_{L^1} := \int_a^b |f(x)|\, dx,$$

that measures the "average" (if you divide by $b - a$) or integral magnitude of a function f. We will see that $\|\cdot\|_{L^1}$ is a norm and therefore induces the distance

$$d_1(f, g) := \|f - g\|_1.$$

The quantity $d_1(f, g)$ measures the average (or integral) discrepancy between f and g. A large pointwise error can have small integral cost when it is confined to a short interval - a feature that may be desired in specific approximation circumstances.

Before testing limits, we verify that $\|\cdot\|_1$ is a norm on continuous functions. Nonnegativity, homogeneity, and the triangle inequality follow from the corresponding properties of the absolute value. The remaining issue is whether zero integral forces a continuous function to vanish everywhere.

Lemma 1.2.1 (Zero integral forces a continuous function to vanish)**.** Let $f \in C([a, b])$. If

$$\int_a^b |f(x)|\, dx = 0,$$

then $f = 0$ on $[a, b]$.

Proof. Suppose that $f(x_0) \neq 0$ for some $x_0 \in [a, b]$. By continuity at x_0, there exists $r > 0$ such that

$$|f(x)| \geq \frac{|f(x_0)|}{2}$$

for every

$$x \in [a, b] \cap (x_0 - r, x_0 + r).$$

This set contains an interval of positive length. Integrating the positive lower bound over that interval gives

$$\int_a^b |f(x)|\, dx > 0,$$

contrary to the hypothesis. Hence f vanishes everywhere. $\square$

We can now ask whether every sequence that should have a limit in this distance actually has one in $C([a, b])$.

Definition 1.2.2 (Complete metric space)**.** A sequence (f_n) in a metric space is **Cauchy** if, for every $\varepsilon > 0$, there exists N such that

$$d(f_m, f_n) < \varepsilon \qquad \forall m, n \geq N.$$

A metric space is **complete** if every Cauchy sequence converges (i.e. has a limit).

However, $C^0([a, b])$ is not complete with respect to the L^1 distance.

Proposition 1.2.3 (Continuous functions are not complete in the integral norm)**.** The normed space

$$(C([-1, 1]), \|\cdot\|_{L^1}), \qquad \|f\|_{L^1} := \int_{-1}^{1} |f(x)|\, dx,$$

is not complete.

Proof. For $n \in \mathbb{N}$, define

$$f_n(x) = \begin{cases} 0, & -1 \leq x \leq 0, \\ 2^n x, & 0 < x < 2^{-n}, \\ 1, & 2^{-n} \leq x \leq 1. \end{cases}$$

Each f_n is continuous.

Let us check that $(f_n)_{n\in\mathbb{N}}$ is Cauchy. If $m \geq n$ it holds that

$$\|f_m - f_n\|_{L^1} \leq 2^{-n}.$$

Indeed, $|f_m(x) - f_n(x)| \leq \mathbb{1}_{[0,2^{-n}]}(x)$.

Now, suppose by contradiction that $f_n \to f$ in L^1 for some $f \in C([-1,1])$. We reason by cases on the value of $f(0)$.

Case $f(0) \neq 0$. By continuity of f at 0, there exists $\delta > 0$ such that $|f(x)| \geq \frac{|f(0)|}{2}$ for every $x \in [-\delta, 0]$. For every n, we have $f_n(x) = 0$ on $[-1, 0]$. Consequently,

$$\|f_n - f\|_{L^1} \geq \int_{-\delta}^{0} |f_n(x) - f(x)|\, dx = \int_{-\delta}^{0} |f(x)|\, dx \geq \delta \frac{|f(0)|}{2} > 0.$$

This contradicts $\|f_n - f\|_{L^1} \to 0$ and thus, necessarily, $f(0) = 0$.

We now use the behavior to the right of 0. Since f is continuous at 0 and $f(0) = 0$, there exists $\delta > 0$ such that $|f(x)| < \frac{1}{2}$ for every $x \in [0, \delta]$. Choose n sufficiently large that $2^{-n} < \frac{\delta}{2}$. For every $x \in [2^{-n}, \delta]$, we have $f_n(x) = 1$, and therefore

$$|f_n(x) - f(x)| = |1 - f(x)| \geq 1 - |f(x)| > \frac{1}{2}.$$

It follows that

$$\|f_n - f\|_{L^1} \geq \int_{2^{-n}}^{\delta} |f_n(x) - f(x)|\, dx > \frac{1}{2}(\delta - 2^{-n}) > \frac{\delta}{4}.$$

This lower bound holds for every sufficiently large n, contradicting the assumption that $\|f_n - f\|_{L^1} \longrightarrow 0$.

Therefore (f_n) has no limit in $C([-1,1])$ with respect to the L^1-norm. Hence $(C([-1,1]), \|\cdot\|_{L^1})$ is not complete. $\square$

One can guess what the missing limit function should be: it is the step function $h = \mathbb{1}_{[0,1]}$. The sequence f_n fails to converge inside $C([-1,1])$ because continuity is not preserved by convergence in L^1.

The step function is Riemann integrable, so one might try to repair the problem by replacing continuous functions with all Riemann-integrable functions.

While a much more subtle argument, we will use Lebesgue theory later in the course to see that Riemann-integrable functions are not complete with respect to the L^1 distance.

Warning. Let $\mathcal{R}([0,1])$ be the vector space of Riemann-integrable functions on $[0,1]$.

For

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0, \end{cases}$$

we have $f \neq 0$ but $\|f\|_1 = 0$.

Thus $\|\cdot\|_1$ is not a norm on pointwise-defined Riemann-integrable functions. To obtain a norm, functions at zero distance must be identified.

Even if we carefully avoid the problem above, some Cauchy sequences of Riemann-integrable functions still fail to converge to a Riemann-integrable limit function.

Proposition (Riemann-integrable functions are incomplete with respect to L^1)**.** There exists a sequence $(f_n)_{n\in\mathbb{N}}$ of continuous functions on $[0,1]$ that is Cauchy with respect to

$$d_1(f,g) := \int_0^1 |f(x) - g(x)|\, dx$$

but there does not exist a Riemann-integrable function h on $[0,1]$ such that $\lim_n d_1(f_n, h) = 0$.

We do not present the proof here. It will be simple once we have developped Lebesgue integration theory.

1.3 The abstract repair: completion

The examples tell us what is wrong with the starting spaces: the integral distance produces Cauchy sequences whose limits are absent. We can use abstract metric completion to "fill in the gaps" and add one new element for each missing limiting behavior.

Definition (Completion of a metric space)**.** A **completion** of a metric space (X,d) is a complete metric space $(\widehat{X}, \hat{d})$ together with an isometric embedding

$$\iota : X \to \widehat{X}$$

such that $\iota(X)$ is dense in $\widehat{X}$.

Theorem (Review: Existence and uniqueness of metric completion)**.** Every metric space has a completion. Any two completions are isometric by an isometry that preserves the embedded copy of the original space.

In the standard construction, an element of $\widehat{X}$ is an equivalence class of Cauchy sequence in X: two Cauchy sequences represent the same element when the distance between their corresponding terms tends to zero.

Applied to $C([a,b])$ with the integral norm, this construction formally supplies all missing limits. We can extend the integral intself to the completion using the existence of continuous extensions.

Proposition (Review: Uniformly continuous maps extend to completions)**.** Let (X, d_X) be a metric space, let $\widehat{X}$ be a completion of X, and let (Y, d_Y) be complete. Every uniformly continuous map

$$T : X \to Y$$

has a unique continuous extension

$$\hat{T} : \widehat{X} \to Y.$$

Define the integration map as

$$I : C([a,b]) \to \mathbb{R}, \qquad I(f) := \int_a^b f(x)\,dx.$$

For $f, g \in C([a,b])$,

$$|I(f) - I(g)| = \left| \int_a^b (f-g)(x)\,dx \right| \leq \int_a^b |f(x) - g(x)|\,dx = \|f - g\|_1.$$

Thus the integral is 1-Lipschitz: functions that are close in integral norm have close integrals. Because I is Lipschitz and $\mathbb{R}$ is complete, I extends uniquely to the completion of $C([a,b])$ in the integral norm. Concretely, if an element of the completion is represented by a Cauchy sequence (f_n), its integral is defined by

$$\hat{I}([(f_n)]) := \lim_{n\to\infty} \int_a^b f_n(x)\,dx.$$

In practice, uniform continuity guarantees both that this limit exists and that equivalent Cauchy sequences give the same value.

This construction accomplishes two goals: it creates a complete space and extends the integral to every new element. It does not yet give a satisfactory description of those elements. Equivalence of L^1 Cauchy sequences of continuous functions is an abstract object; from that description alone, it is not even clear whether the new element is representable as a "function".

The measure-theoretic construction provides a concrete model for this abstract completion. After Lebesgue measure and integration have been developed, the completion above can be identified with the *Lebesgue* space $L^1([a,b])$.

1.4 An important example: the fat rationals

We recall two elementary topological notions.

Definition (Open and dense subsets)**.** Let (X, d) be a metric space. A set $E \subset X$ is **open** if for every $x \in E$ there exists $r_x > 0$ s.t. $B_{r_x}(x) \subset E$.

The set $E \subset X$ is **dense** if any one of the following equivalent conditions holds:

- its closure $\bar{E}$ is X;
- every open subset of $A \subset X$ intersects E.

We henceforth reason on the metric space $[0, 1]$ with the standard Euclidean distance. A set can be both open and dense. For example, the sets $(0, 1)$ or $[0, 1/2) \cup (1/2, 1]$ are open and dense in $[0, 1]$.

We now give a different construction that also controls the total length of the intervals used to form the open set. List all rational numbers in $[0, 1]$, allowing repetitions, by successively listing the fractions with each positive denominator:

$$\begin{aligned}
&q_0 = \frac{0}{1}, \qquad q_1 = \frac{1}{1}, \\
&q_2 = \frac{0}{2}, \qquad q_3 = \frac{1}{2}, \qquad q_4 = \frac{2}{2}, \\
&q_5 = \frac{0}{3}, \qquad q_6 = \frac{1}{3}, \qquad q_7 = \frac{2}{3}, \qquad q_8 = \frac{3}{3}, \qquad \ldots
\end{aligned}$$

Every rational number in $[0, 1]$ appears somewhere in this sequence. Thus $\mathbb{Q} \cap [0, 1] = \{q_n : n \geq 0\}$.

Definition 1.4.1 (Fat rationals on $[0, 1]$)**.** Recall that on $[0, 1]$ we have $B_\rho(x) := (x - \rho, x + \rho) \cap [0, 1]$.

Fix $r > 0$ and define

$$F_r := [0, 1] \cap \bigcup_{n=0}^{\infty} B_{r2^{-n}}(q_n).$$

We refer to F_r informally as the set of *fat rationals*.

Each rational q_n has been replaced by an interval whose radius decreases with n. The set F_r is open in $[0, 1]$ since it is a union of open sets. It contains every rational point of $[0, 1]$, since $q_n \in B_{r2^{-n}}(q_n)$ for every n. Because the rational numbers are dense, F_r is dense in $[0, 1]$.

Intuitively, the intervals in the definition of F_r have total length

$$\sum_{n=0}^{\infty} 2r2^{-n} = 4r.$$

Therefore, when r is sufficiently small, these intervals cannot cover all of $[0,1]$. Next, we make this statement precise.

Proposition 1.4.2 (The fat rationals are a proper subset of $[0,1]$)**.** If $0 < r \leq 1/4$, then F_r is a proper of $[0,1]$.

Proof. Suppose, to the contrary, that $[0,1] = \bigcup_{n=0}^{\infty} B_{r2^{-n}}(q_n)$, i.e. $\left\{B_{r2^{-n}}(q_n)\right\}_{n\in\mathbb{N}}$ is an open cover of the compact interval $[0,1]$. By compactness, there exists a finite set of indices $\mathcal{J}$ such that

$$[0,1] \subseteq \bigcup_{n\in\mathcal{J}} B_{r2^{-n}}(q_n).$$

For $n \in \mathcal{J}$, set

$$a_n := q_n - r2^{-n}, \qquad b_n := q_n + r2^{-n}.$$

so that $B_{r2^{-n}}(q_n) = (a_n, b_n) \cap [0,1]$.

Next, we extract a chain of overlapping intervals from the finite cover. Since 0 is covered, there exists $n_0 \in \mathcal{J}$ such that $a_{n_0} < 0 < b_{n_0}$ If $b_{n_0} > 1$, the chain already covers the interval from a point to the left of 0 to a point to the right of 1.

Otherwise, $b_{n_0} \in [0,1]$. Since b_{n_0} is covered by the finite collection of open intervals, there exists $n_1 \in \mathcal{J} \setminus \{n_0\}$ such that $a_{n_1} < b_{n_0} < b_{n_1}$. In particular, $b_{n_1} > b_{n_0}$. If $b_{n_1} \leq 1$, we repeat the argument. At the k-th step, choose an index $n_{k+1} \in \mathcal{J} \setminus \{n_0, \dots n_k\}$ such that

$$a_{n_{k+1}} < b_{n_k} < b_{n_{k+1}}.$$

The indices chosen at successive steps are distinct by construction and since $\mathcal{J}$ is finite the process terminates. We therefore obtain indices $n_0, n_1, \dots, n_M \in \mathcal{J}$ such that

$$a_{n_0} < 0, \qquad a_{n_k} < b_{n_{k-1}} < b_{n_k} \quad \text{for } 1 \leq k \leq M, \qquad b_{n_M} > 1.$$

It follows that

$$1 < b_{n_M} - a_{n_0}.$$

On the other hand, the overlap relations give

$$\begin{aligned}
b_{n_M} - a_{n_0} &= (b_{n_0} - a_{n_0}) + \sum_{k=1}^{M} (b_{n_k} - b_{n_{k-1}}) \\
&< (b_{n_0} - a_{n_0}) + \sum_{k=1}^{M} (b_{n_k} - a_{n_k}) \\
&= \sum_{k=0}^{M} (b_{n_k} - a_{n_k}) \\
&\leq \sum_{n \in \mathcal{J}} (b_n - a_n) \\
&\leq \sum_{n=0}^{\infty} 2r2^{-n} = 4r.
\end{aligned}$$

Consequently $1 < 4r$, which is a contradiction to our assumption $r \leq 1/4$. Therefore $F_r \neq [0,1]$. $\square$

Problem. Identify precisely where compactness and finiteness are used in the proof of Proposition 1.4.2. Explain why the overlapping-chain argument cannot be used applied directly to $\{B_{r2^{-n}}(q_n)\}_{n\in\mathbb{N}}$

1.4.1 Incompleteness of Riemann integrable functions

The construction of the fat rationals can be used to prove that Riemann integrable functions do not form a complete space with the L^1 distance. We provide a sketch of the argument

Fix $0 < r < 1/4$, and define

$$F_n := [0,1] \cap \bigcup_{k=0}^{n} B_{r2^{-k}}(q_k).$$

and $F = \bigcup_n F_n$. We show next that $(F_n)_(n \in \mathbb{N})$ is Cauchy in L^1.

Each F_n is a finite union of intervals, so f_n has only finitely many discontinuities and is Riemann integrable. It holds that

$$\|f_m - f_n\| = \int_0^1 |f_m(x) - f_n(x)|\, dx \leq \sum_{k=n+1}^{m} 2r2^{-k} \leq 2r2^{-n}.$$

since the sets F_n are increasing and therefore

$$0 \leq f_m - f_n \leq \sum_{k=n+1}^{m} \mathbb{1}_{B_{r2^{-k}}(q_k)\cap[0,1]}.$$

This shows that (f_n) is Cauchy with respect to d_1.

Next, let us argue that $\mathbb{1}_F$ is not Riemann integrable. This does not conclude the proof but provides an argument for it, conditional on showing that the L^1 limit of the F_N must essentially be $\mathbb{1}_F$.

Since F contains $\mathbb{Q} \cap [0,1]$, it is dense in $[0,1]$. Hence every nondegenerate subinterval of $[0,1]$ intersects F, and therefore $U(\mathbb{1}_F; \tau) = 1$ for every partition τ. It remains to bound the lower sums. Let

$$\tau = (\tau_0, \dots, \tau_{\#\tau})$$

be a partition, and let $\mathcal{J}_\tau := \{j \in \{0, \dots, \#\tau - 1\} : [\tau_j, \tau_{j+1}] \subseteq F\}$. There are the indexes of the intervals of the partition where $\inf f(x) = 1$ and not $\inf f(x) = 0$. More precisely, since $\mathbb{1}_F$ takes only the values 0 and 1 we have $L(\mathbb{1}_F; \tau) = \sum_{j \in \mathcal{J}_\tau} (\tau_{j+1} - \tau_j)$. The set $K_\tau := \bigcup_{j \in \mathcal{J}_\tau} [\tau_j, \tau_{j+1}]$ is a compact subset of F. Consequently, it is covered by finitely many of the intervals $B_{r2^{-k}}(q_k)$. By the same finite-cover estimate used above,

$$L(\mathbb{1}_F; \tau) \leq \sum_{k=0}^{\infty} 2r2^{-k} = 4r.$$

Thus every upper sum equals 1, whereas every lower sum is at most $4r < 1$. Hence $\mathbb{1}_F$ is not Riemann integrable.

Finally, it remains to rule out the possibility that $(f_n)_{n \in \mathbb{N}}$ converges in L^1 to some other Riemann-integrable function. This is technically involved argument that we omit here.